

Leak Diagnosis Process — Species Physics and a Decision Tree

July 31, 2026 · Companion to the pump-down simulator and dewar_vacuum_model. All worked numbers use the project's representative hardware (0.1 L sealed dewar, 10^{-3} Torr end-of-life; pumped-system examples use the nXDS10iC + nEXT85 station and the 10 mm × 10 cm bottleneck, $S_{\text{eff}} \approx 1.05$ L/s at base).

1. Why molecular mass (AMU) matters — the species matrix

Every transport and pumping process in the system is mass-selective, and the *pattern* of that selectivity is the diagnostic toolkit. At 295 K:

Gas	M (amu)	C_molecular / C(N ₂) — $\propto \sqrt{(28/M)}$	Viscosity (μPa·s)	C_viscous / C(N ₂) — $\propto 1/\eta$	Free-molecular heat factor vs N ₂	80 K cryopumped?	Gettered?
H ₂	2	3.73	8.9	2.00	3.65	no	yes
He	4	2.65	19.8	0.90	1.76	no	no
H ₂ O	18	1.25	9.9	1.80	1.47	yes	yes
N ₂	28	1.00	17.8	1.00	1.00	no	yes (slow)
O ₂	32	0.94	20.6	0.86	0.94	no	yes
Ar	40	0.84	22.6	0.79	0.56	no	no
CO ₂	44	0.80	14.9	1.19	1.02	yes	yes

Four readings of this table:

Bottlenecks (molecular regime). Conductance is $\propto$ mean thermal speed $\propto \sqrt{(T/M)}$ — nothing else about the molecule matters once flow is molecular. Hydrogen clears the same tube 3.7× faster than nitrogen; CO₂ 20 % slower; the full H₂→CO₂ span is ×4.7. This is why the simulator scales the water species' molecular conductance by $\sqrt{(29/18)} = 1.27$.

Bottlenecks (viscous regime). Mass drops out; *viscosity* rules — and the ordering scrambles. Helium is actually slightly *worse* than nitrogen viscously ($\eta_{\text{He}} > \eta_{\text{N}_2}$, a monatomic-gas quirk), while H₂ and H₂O flow ~2× and ~1.8× better. Practical consequence: species distinctions barely matter above ~1 Torr and matter enormously below 10^{-3} Torr.

Pumps invert the trend. A turbo's blades must outrun the molecule's thermal speed, so the *fast* gases are the *hard* gases: the nEXT85's speed only sags mildly for He and H₂ (~80 and ~60 L/s vs 84 for N₂) but its compression ratio collapses from $>10^{11}$ (N₂) to $\sim 10^7$ (He) to $\sim 2 \times 10^5$ (H₂) — which is why the residual gas at a turbo's ultimate pressure is hydrogen, and why He backstreaming limits sensitivity in leak detectors. The 80 K cryopump inside an operating dewar is the most selective pump of all: perfect for H₂O/CO₂, zero for everything lighter than CO₂ except nothing. And the getter pumps everything chemically active but is blind to He, Ar, CH₄.

Do we need to track species separately? For *pump-down time*: no — it's air-dominated, and the ± 25 % water-conductance correction is already in the model. For *diagnosis*: absolutely — every

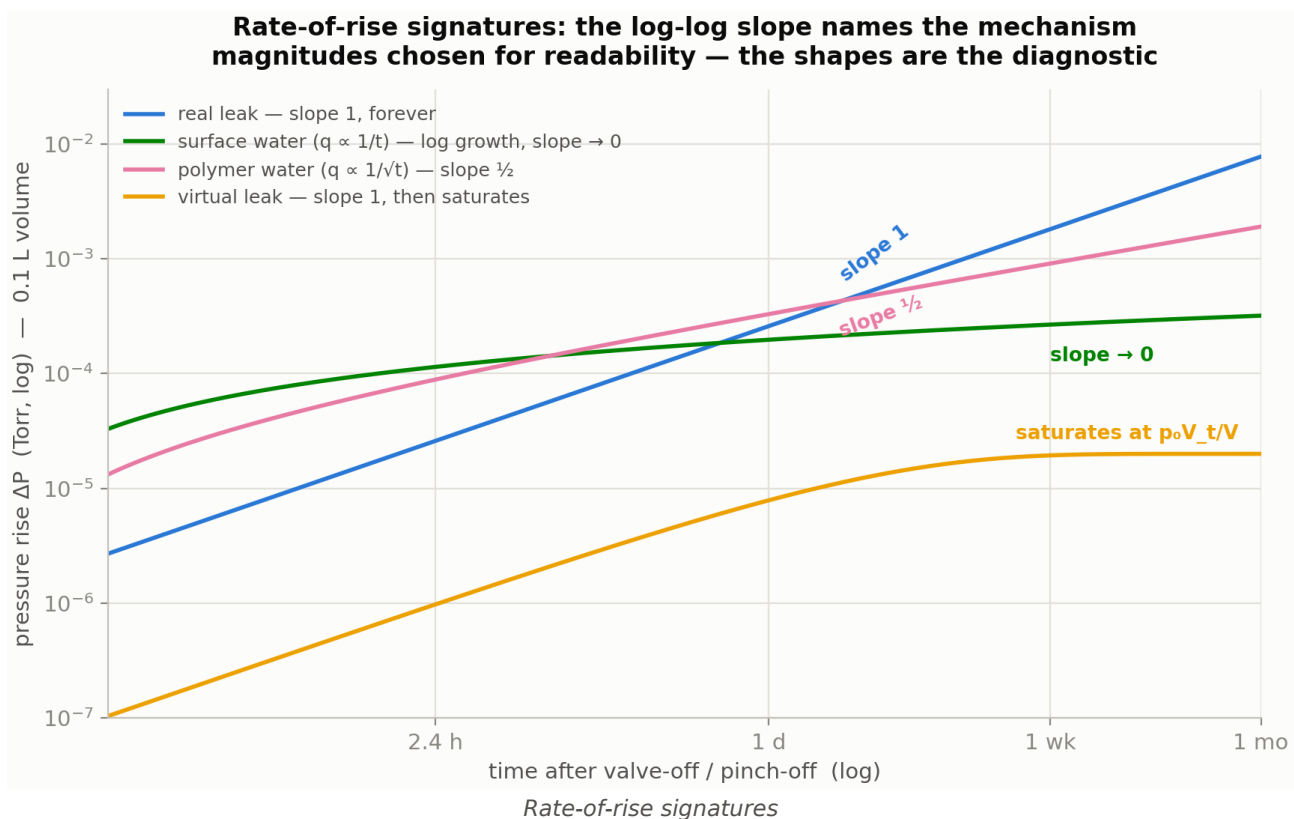
test below works precisely because mass separates the species. Helium is the probe gas of choice for exactly four properties: light (fast conductance, fast response), rare (5.2 ppm atmospheric background), inert, and invisible to getters and cryosurfaces (a He signal cannot be pumped away or confused with outgassing).

Leak-rate unit conversions (same channel, molecular flow): $Q_{\text{air}} = Q_{\text{He}} \cdot \sqrt{4/29} = \mathbf{0.372 \times Q_{\text{He}}}$; $Q_{\text{He}} = 2.69 \times Q_{\text{air}}$. In viscous (gross) leaks the ratio is η -based and ≈ 1.1 — near unity. Always state which convention a spec uses.

2. The decision tree

Symptom → Step 1: does the base pressure floor *fall* or *hold*? (pumped system, days timescale — or the simulator's leak toggle). A floor that decays, however slowly, is outgassing. A floor that holds constant indefinitely is a leak or permeation: $P_{\text{floor}} = Q/S_{\text{eff}}$. Worked: 10^{-6} Torr·L/s of air through our bottleneck station floors at 9.5×10^{-7} Torr — measured and predicted agree in the simulator to three digits.

Step 2: valve off (or pinch off) and read the rate-of-rise *shape*, not just its size. On a log-log plot of ΔP vs time the slope names the mechanism:



log-log slope of $\Delta P(t)$	Mechanism	Check
1, forever	real leak or permeation	$\Delta P(2t)/\Delta P(t) = 2.0$ exactly; composition is air-like
$\frac{1}{2}$	polymer/adhesive water (diffusion)	rises with recent thermal history; m/z 18

→ 0 (logarithmic)	surface water desorption	shrinks decade-per-decade of prior pumping time
1 early, then hard saturation	virtual leak (trapped volume)	total $\Delta P \cdot V \approx p_0 V_t$ — a <i>finite</i> inventory; $\tau = V_t/C$ of weeks

The two-point test is the cheapest discriminator in vacuum practice: measure dP/dt at t and at $3t$ after valve-off. Ratio 1.0 → leak. Ratio $\sim 1/3$ → surface water. Ratio ~ 0.58 ($1/\sqrt{3}$) → diffusion.

Step 3: composition (residual gas analyzer). Air has a fingerprint no outgassing can fake:

Signature	Reading
m/z 28 : 32 $\approx$ 4 : 1, with m/z 40 (Ar) at $\sim 1\%$ of 28, m/z 14 present	air leak — the O_2 and Ar are the tell; outgassing CO also sits at 28 but brings no 32 and no 40
m/z 18 dominant with 17 at $\sim 23\%$ of 18, plus 2 and 44	outgassing (water, hydrogen, CO_2) — the healthy unbaked spectrum
m/z 28 large, 32 absent	CO from surfaces/hot filaments, not a leak
m/z 4 above background	helium — see Step 4; in a <i>sealed gettered</i> dewar any He/Ar growth is damning, since nothing inside makes or pumps them

Step 4: helium testing, matched to the article. For a *pumped* article: spray-probe with a mass-spectrometer leak detector or the RGA on m/z 4, working top-down and upwind, with response time $\tau = V/S_{He}$ per event (≈ 35 s for our 100 L chamber — wait it out or you mislocate the leak). Sensitivity $\sim 10^{-9}$ – 10^{-10} atm·cc/s routinely. For a *sealed* article (pinch-off dewar): bombing (pressurize in He, then detect efflux, MIL-STD-883 TM1014-style) for gross-to-fine screening, or the highest-sensitivity option, **accumulation**: park the article in a small purged enclosure and let helium (or, for a sealed dewar's interior, argon from an air leak) integrate for hours-to-weeks; $Q = \Delta P \cdot V_{enclosure}/t$. Accumulation methods reach 10^{-13} atm·cc/s and, in laboratory practice with RGA sampling, $\sim 10^{-18}$ mbar·L/s for noble gases (Chiggiato) — background control and purge quality, not the instrument, set the floor.

Step 5: judge against the requirement, not the test floor. From the project's budget math: a 10-year ungettered dewar allows 4×10^{-13} atm·cc/s total — below any spray test; a gettered dewar's argon-accumulation limit allows 4.5×10^{-11} atm·cc/s of air — still below routine spray sensitivity. So for sealed dewars the spray/bomb test is a **gross-defect screen**; life-grade hermeticity is demonstrated by process control plus the accumulation-class tests and, in the fleet, by trending warm rate-of-rise and cooldown time across storage intervals — a linear-in-time trend line is a leak; a decelerating one is outgassing.

3. The process, condensed

1. Trend the pumped base pressure (or sealed ROR) — **falling = outgassing, constant = leak**.
2. Valve off; log-log the rise; apply the two-point slope test (1 vs $1/3$ vs $1/\sqrt{3}$; saturation = virtual leak).
3. RGA: look for the O_2 +Ar air fingerprint; He/Ar growth in a sealed gettered volume is a leak, full stop.
4. Localize with He spray (mind $\tau = V/S$); quantify sealed articles by accumulation; convert He↔air by $\times 0.372$ (molecular).

5. Compare to the budget-derived allowable, and record the result as a *rate*, stating the gas and conversion.

Sources: species table computed from kinetic theory ($\sqrt{28/M}$) and standard 295 K viscosities; turbo per-gas speeds and compressions from the Edwards nEXT brochure; accumulation sensitivity from Chiggiato, CERN Yellow Rep. arXiv:2006.07124; He↔air conversions are exact molecular-flow kinematics; MIL-STD-883 TM1014 named for the bombing method's lineage. Budget allowables derived in vacuum_degradation_sources.md and dewar_vacuum_model/REVIEW.md.